

Bridging the Perception–Reasoning Gap for Long–Term Robot Navigation

– and the Research Journey

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2026.07.23



Giseop Kim

Team Leader



Giseop Kim

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[Google Scholar](#)

Biography

I am an Assistant Professor in the Department of Robotics and Mechatronics Engineering at DGIST, where I lead the Autonomy and Perceptual Robotics Lab (APRL). I also hold joint appointments in the Department of Artificial Intelligence and the Mechanical Engineering Track at DGIST.

Before joining DGIST, I was a Research Scientist at NAVER LABS (2021-2024), working on autonomous driving and 3D vision. I received my PhD, MS, and BS from KAIST, all in Civil and Environmental Engineering, under the guidance of Prof. Ayoung Kim.

My research focuses on enabling robots to perceive and navigate the real world robustly and efficiently through SLAM, 3D reconstruction, sensor fusion, and spatial AI.

[Research Statement](#) | [Teaching Statement](#)

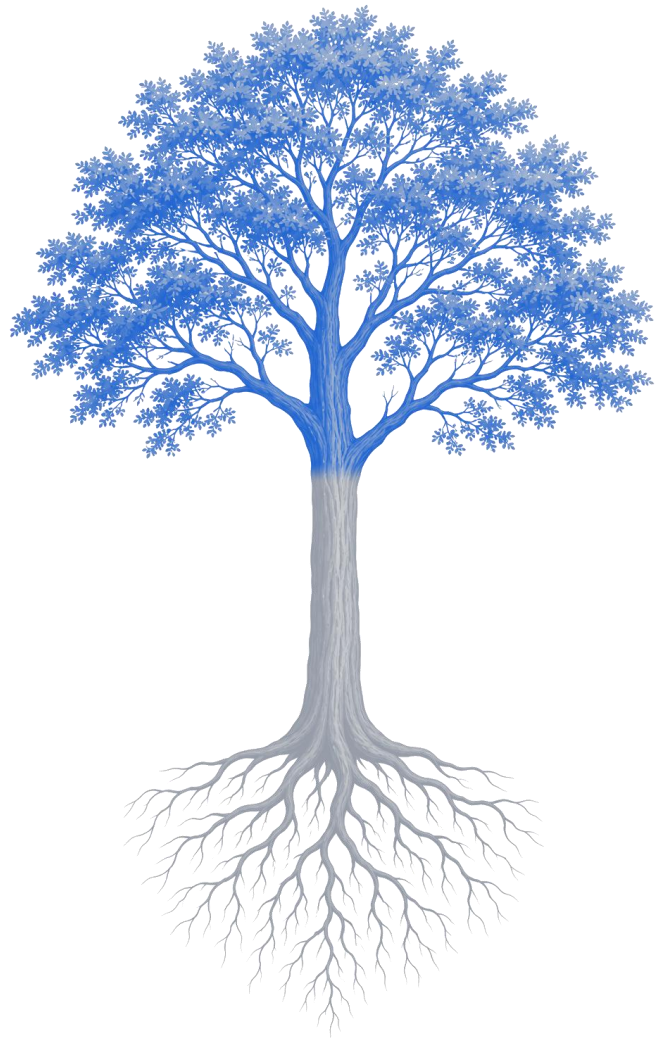
Research Interests

- Simultaneous Localization and Mapping (SLAM)
- Mobile Robot Navigation
- Spatial AI & Physical AI

Education

- PhD in Civil & Env. Eng., 2022, KAIST
- MS in Civil & Env. Eng., 2019, KAIST
- BS in Civil & Env. Eng., 2017, KAIST

Chapter 0: Future



Visual Language Action (VLA) model



Visual Language Action (VLA) model for Navigation



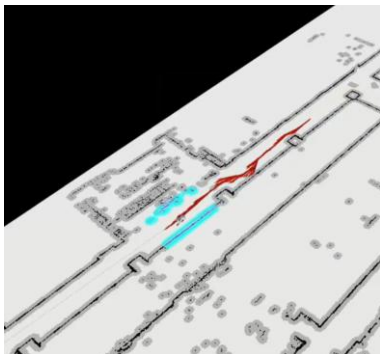
What, Why, When?



What is Visual-Language Navigation?

- “Where is the elevator?”

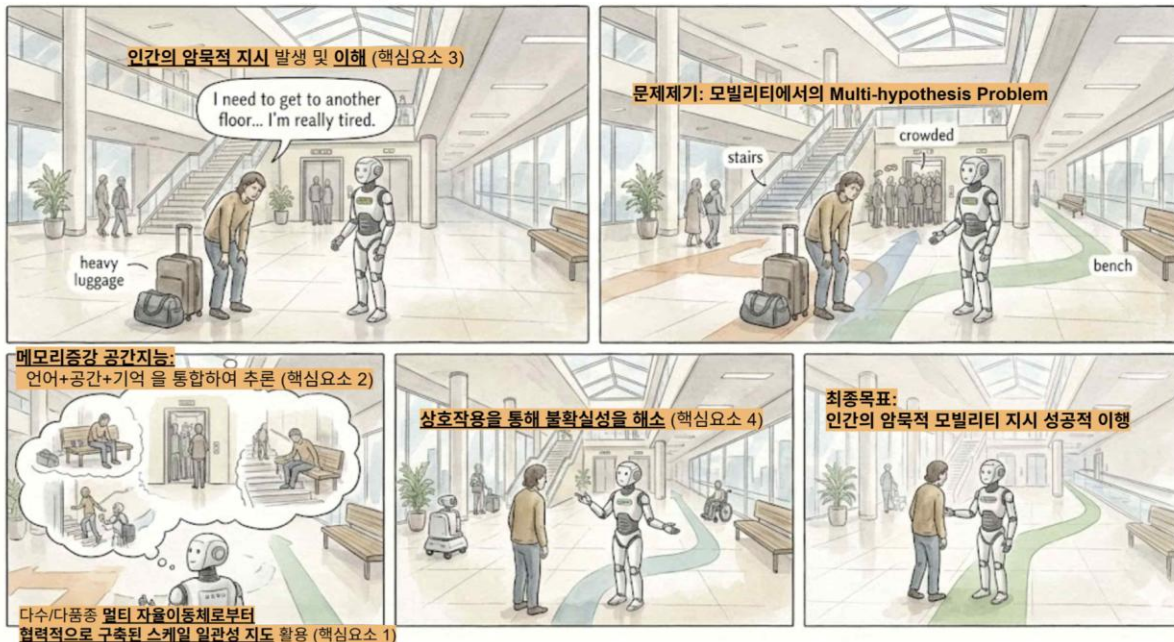
```
=====
AI Navigation Client 시작 (Server: http://
=====
명령 입력 (종료: exit): Where is a elevator?
서버 분석 중...
DEBUG: {'type': 'position', 'text': 'A elevator is
'position': [11.848, -11.848, 0.0], 'orientation':
```



When Visual-Language Navigation?

- Human-robot Interactive Navigation (HRIN)

Memory-Augmented Spatial AI for Autonomous Mobility



Why Visual-Language Navigation?

- Robot Web Era



Apple iPhone LiDAR RoomPlan



Project Aria from Meta



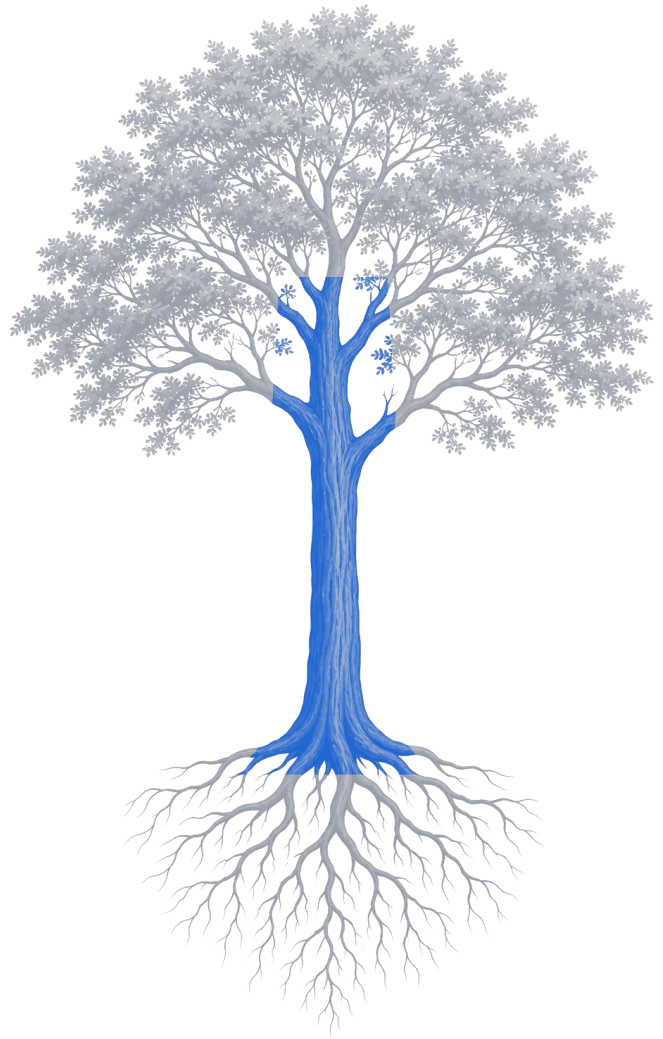
Unitree Go1 Legged Robot



NAVER LABS Unmanned Shuttle (ALT-B)

- DGIST 컨실리언스홀 전체를 이종로봇 5대가 자율적으로 돌아다니며 하루 안에 공간적으로 이해하려면 어떻게 해야할까요?
- 휴머노이드 로봇이 대중교통을 이용해 강남에서 DGIST까지 안전하게 오려면 어떤 기술이 필요할까요?
- 네비게이션의 플래닝 및 의사결정에서 사람의 언어적 및 비언어적 표현은 어떻게 통합해야 할까요?

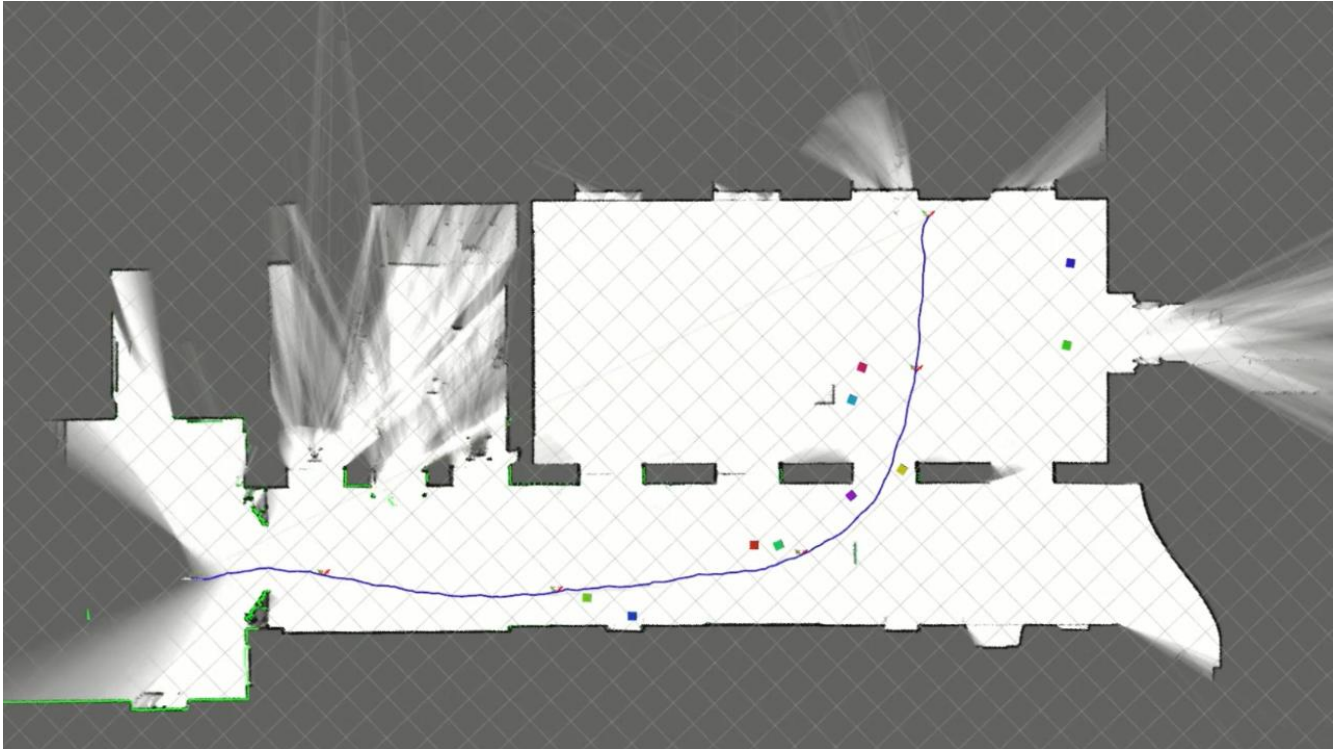
Chapter 1: Present



SLAM (Simultaneous Localization and Mapping)



SLAM (Simultaneous Localization and Mapping)



Google Cartographer, 2016

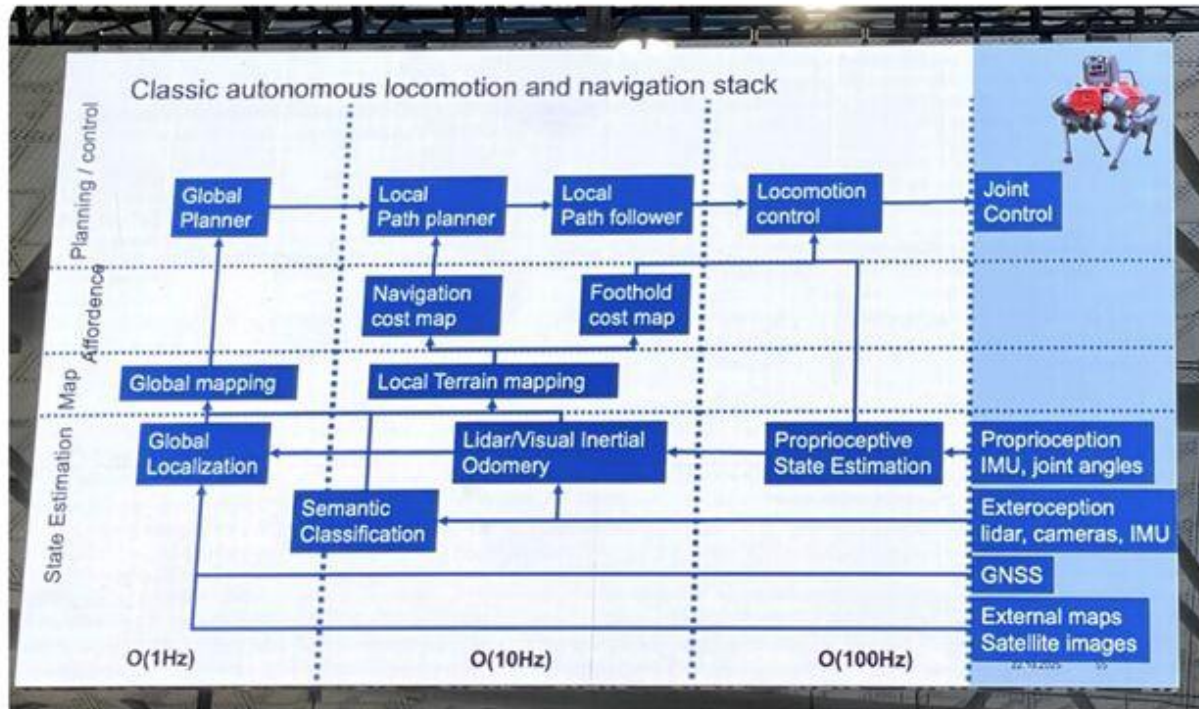
SLAM (Simultaneous Localization and Mapping)



Autonomous Robot Navigation



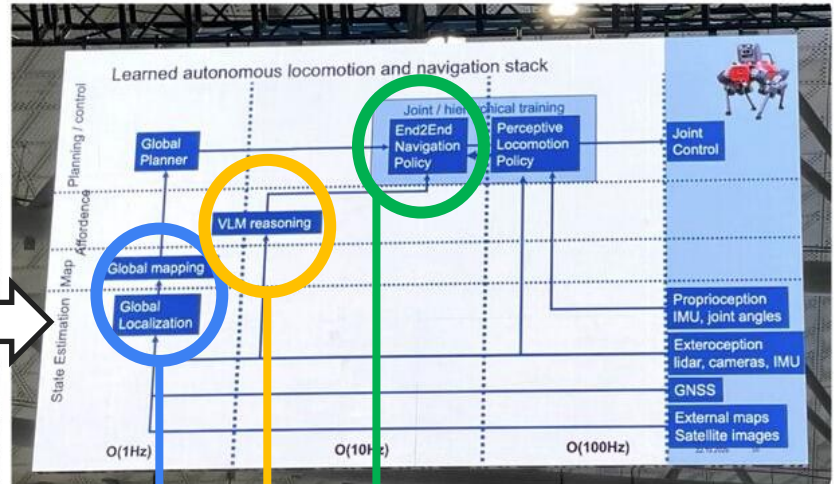
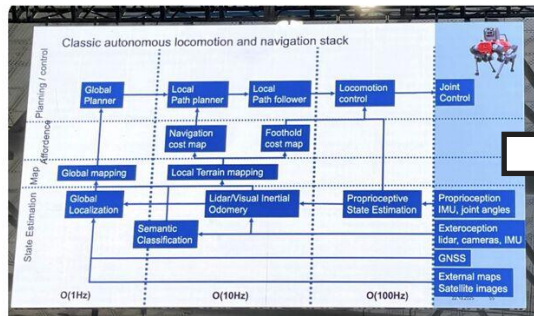
What is Visual-Language Navigation?



Autonomous Robot Navigation

– Historical View (a.k.a PPP)

Source: Prof. Marco Hutter,
from IROS2025



P1: Perception

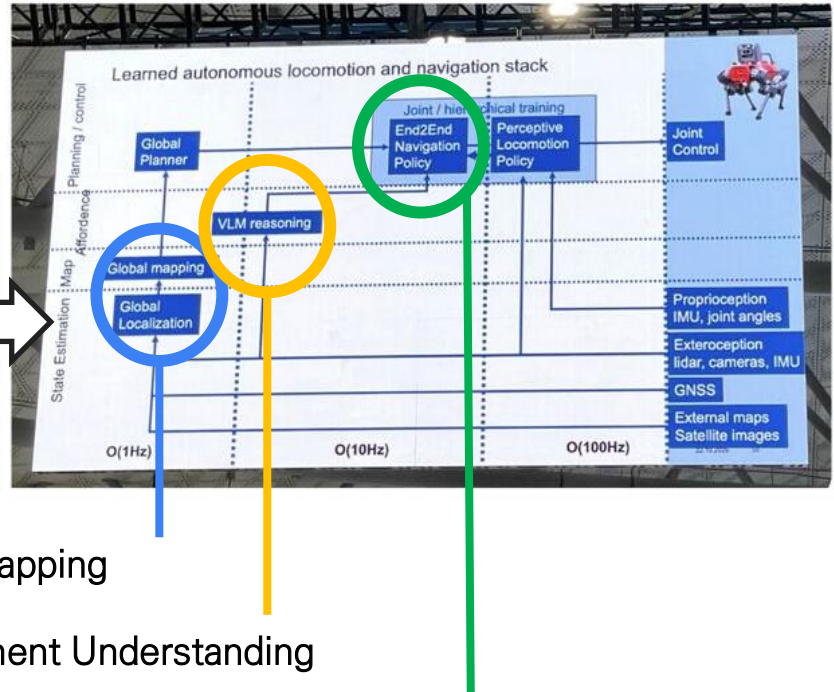
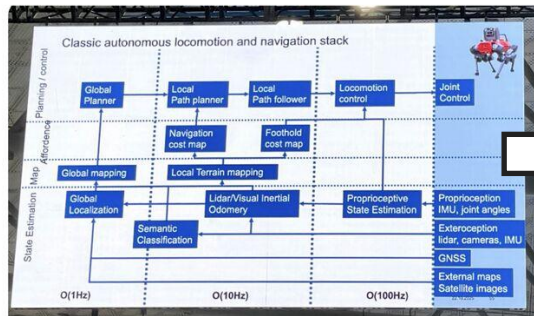
P2: (P1+) Prediction

P3: (P1+P2+) Planning

Autonomous Robot Navigation

– APRL Perspective (Task-centric)

Source: Prof. Marco Hutter,
from IROS2025



Human-Robot Collaborative Mapping

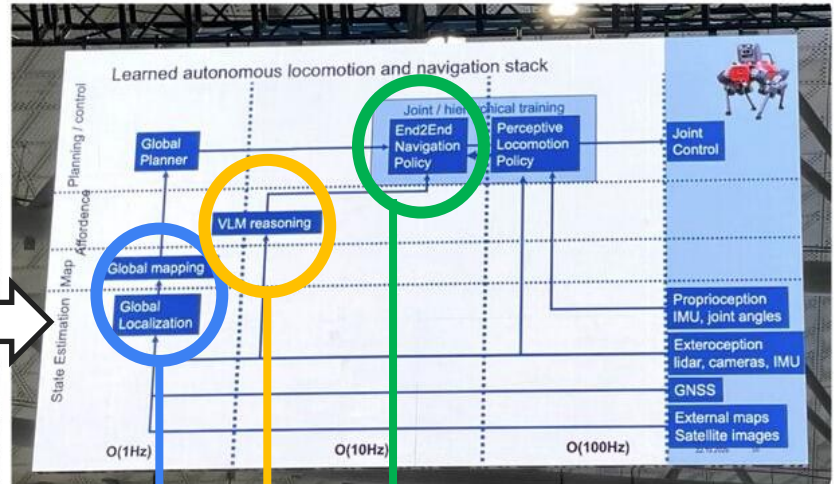
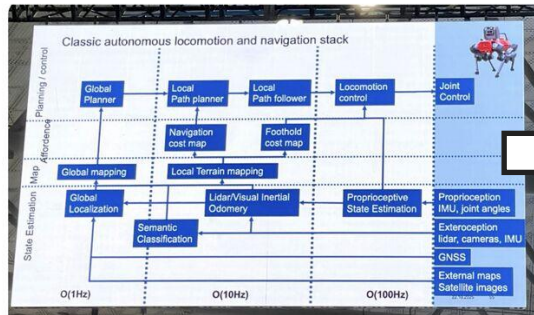
Continual Human-Environment Understanding

Human-Robot Interactive Navigation

Autonomous Robot Navigation

– APRL Perspective (Method-centric)

Source: Prof. Marco Hutter,
from IROS2025



Geometric Foundation Models

Map and Memory Management

Visual-Language Navigation

Section 1

Human-Robot Collaborative Mapping in Robot Web Era

Geometric Foundation Models

DUST3R: Geometric 3D Vision Made Easy

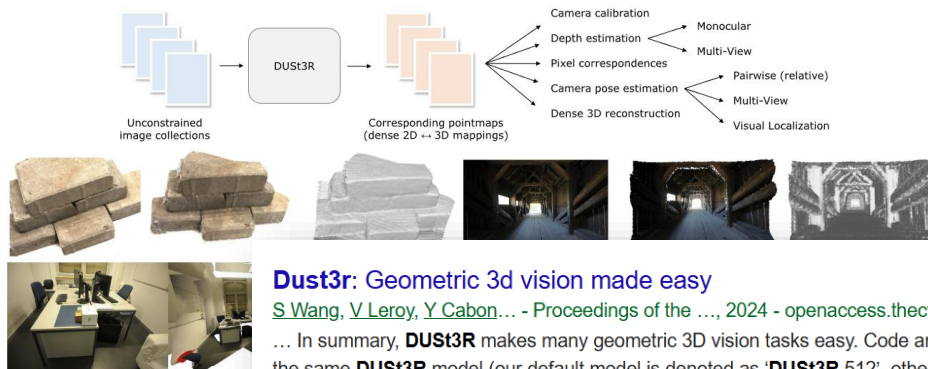
Shuzhe Wang*, Vincent Leroy†, Yohann Cabon†, Boris Chidlovskii† and Jerome Revaud†

*Aalto University

†Naver Labs Europe

shuzhe.wang@aalto.fi

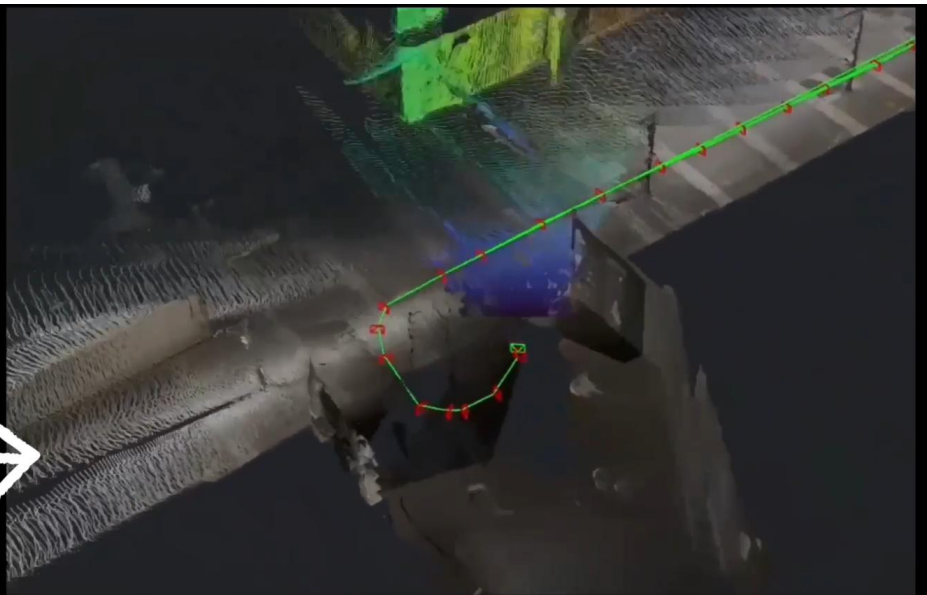
firstname.lastname@naverlabs.com



- Wang, Shuzhe, et al. "Dust3r: Geometric 3d vision made easy." CVPR 2024
- Murai, Riku, Eric Dexheimer, and Andrew J. Davison. "Mast3r-slam: Real-time dense slam with 3d reconstruction priors.", CVPR 2025

Geometric Foundation Models

Visual SLAM
at Consilience Hall
(E7), DGIST

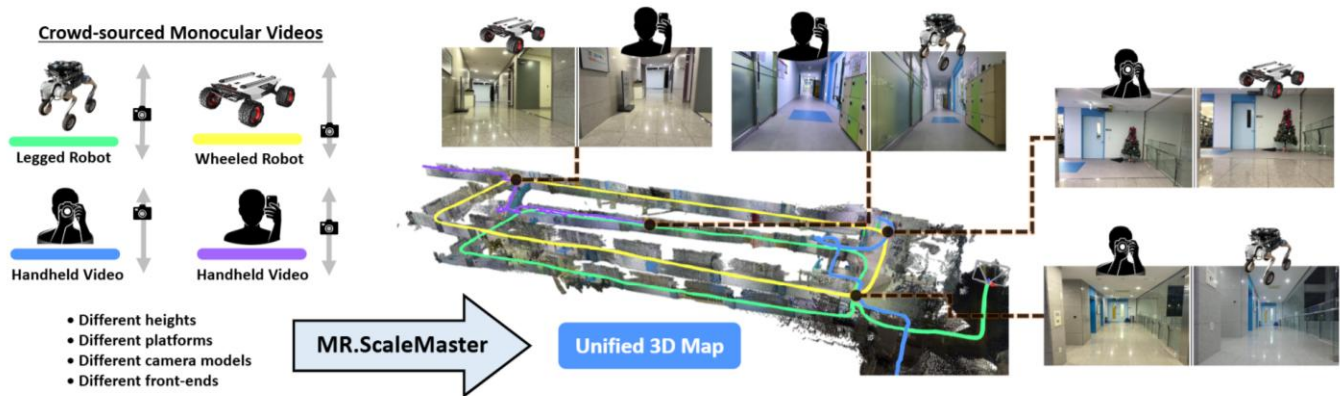


Hyoseok Ju, Bokeon Suh, and Giseop Kim. "Have We Mastered Scale in Deep Monocular Visual SLAM? The ScaleMaster Dataset and Benchmark.", ICRA 2026



Human-Robot Collaborative Mapping

- In the GFM era.





Human-Robot Collaborative Mapping

Real-World Multi-Robot Dense Mapping

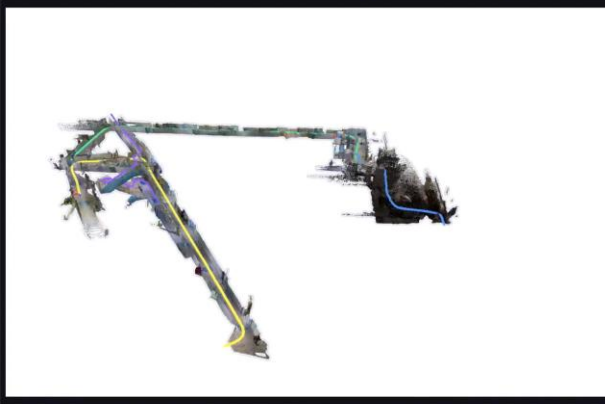
Multi-floor indoor environment · 4 heterogeneous agents

AGENTS

- Legged Robot
- Handheld 1
- Wheeled Robot
- Handheld 2



Data Collection



Corridor Environment

HIGHLIGHTS

4 heterogeneous agents fused into single map

Inter-agent loop closures across different platforms

No backend modification per front-end

Hyoseok Ju, and Giseop Kim. "MR.ScaleMaster: Scale-Consistent Collaborative Mapping from Crowd-Sourced Monocular Videos." arXiv preprint arXiv:2604.11372 (2026).



KITTI 00,

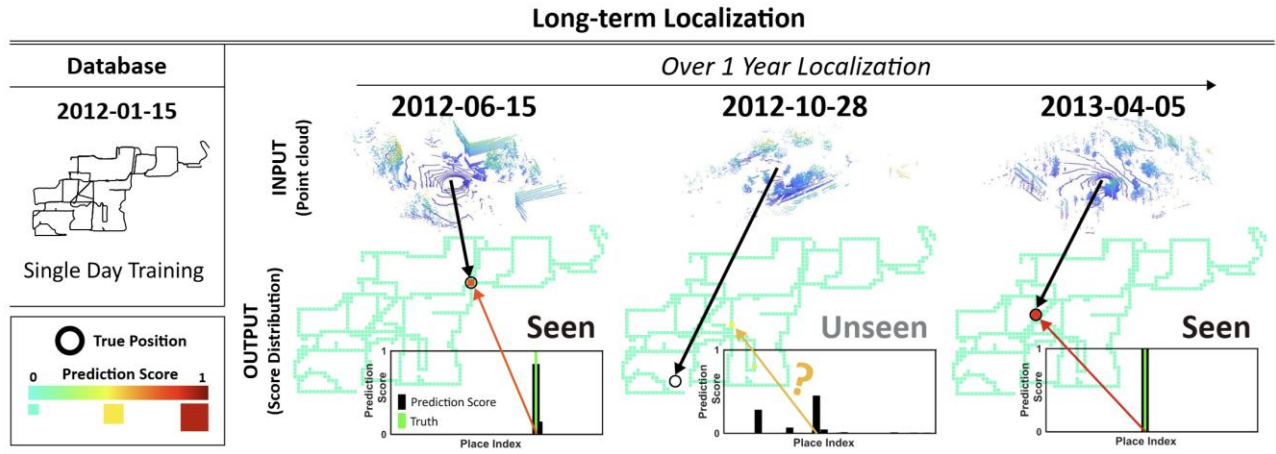
MR.ScaleMaster
with MAST3R-SLAM

Section 2

Continual Human–Environment Understanding

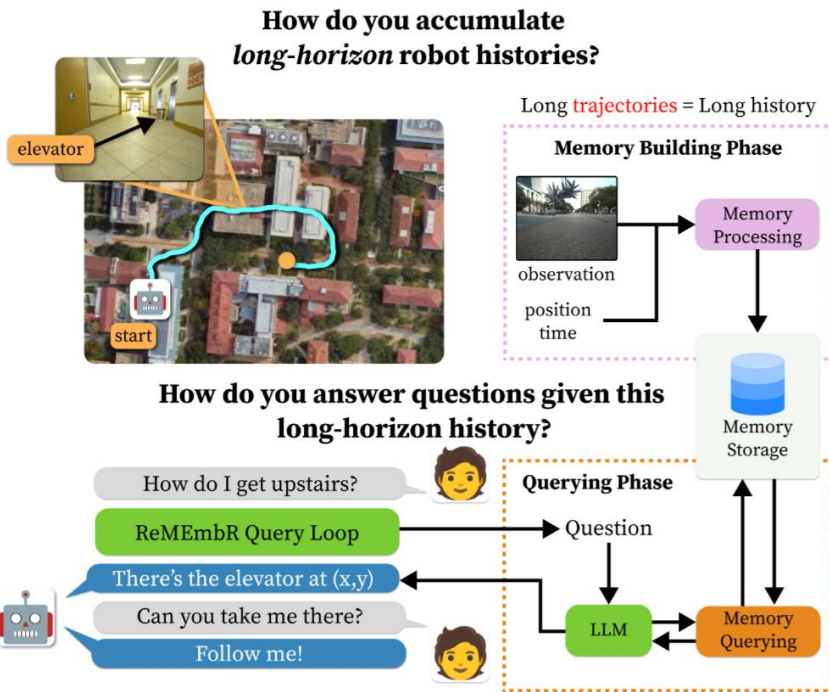
Traditional Lifelong Navigation

- Map for Long-term Navigation



Giseop Kim, Byungjae Park, and Ayoung Kim. "1-day learning, 1-year localization: Long-term lidar localization using scan context image." RA-L (2019)

Memory for Robot Navigation





Memory for Lifelong Robot Navigation

- Long-term Navigation Memory

Example Query

“When did the scissor first appear?”
“Was the vacuum in the room at Session 9?”
“Did the brown basket move at Session 6?”
“Did the fridge stay in place at Session 4?”
“Did the green chair come back at Session 9?”

“Where has the blue totebag been across all sessions?”
“Which object moved most frequently?”
“When was the last time the white board moved?”
“Was the robot dog in the same location in S_1 and S_{10} ?”



Memory for Lifelong Robot Navigation

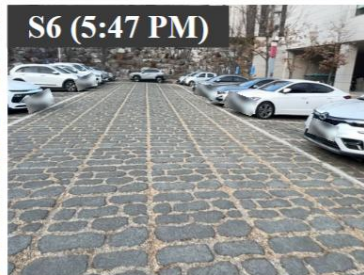
- Long-term Navigation Memory





Memory for Lifelong Robot Navigation

- VQA for Lifelong Scene Understanding



Q: When should I go if I want to find a parking spot?

A: Come around 8:00AM to find a parking spot (Session3, 0 cars at that time).

Q: When is it hardest to find a place to park?

A: It's hardest to find a spot around 1:03PM (Session8, 34 cars).

Avoid that time if you want an empty spot.

Secion 3

Human-Robot Interactive Navigation

VLA

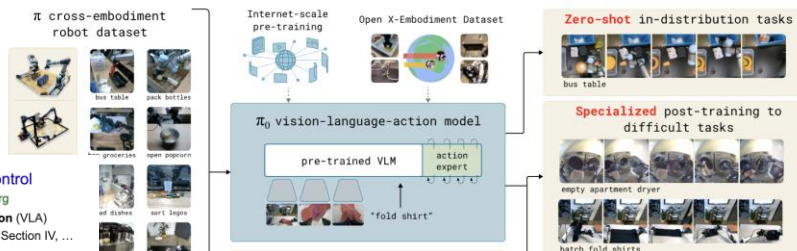
• π_0

π_0 : A Vision-Language-Action Flow Model for General Robot Control

Physical Intelligence

Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoyang Shi, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, Ury Zhilinsky
<https://physicalintelligence.company/blog/pi0>

31 Oct 2024



π_0 : A Vision-Language-Action Flow Model for General Robot Control

K Black, N Brown, D Driess, A Esmail, M Equi, ... - arXiv preprint arXiv ..., 2024 - arxiv.org

... models. Our work is most closely related to recently proposed **visionlanguage action (VLA)**

models. ... We compare this fine-tuned **π_0 model** with the **π_0 small model** described in Section IV, ...

☆ 저장 50 인용 1898회 인용 관련 학술자료 전체 4개의 버전 80

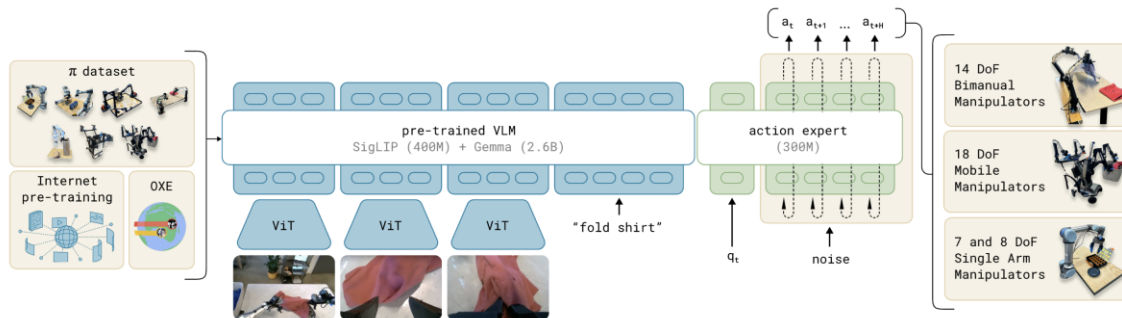
$\pi_{0.5}$: a Vision-Language-Action Model with Open-World Generalization

K Black, N Brown, J Dargatzidis

... We describe **$\pi_{0.5}$** , a new

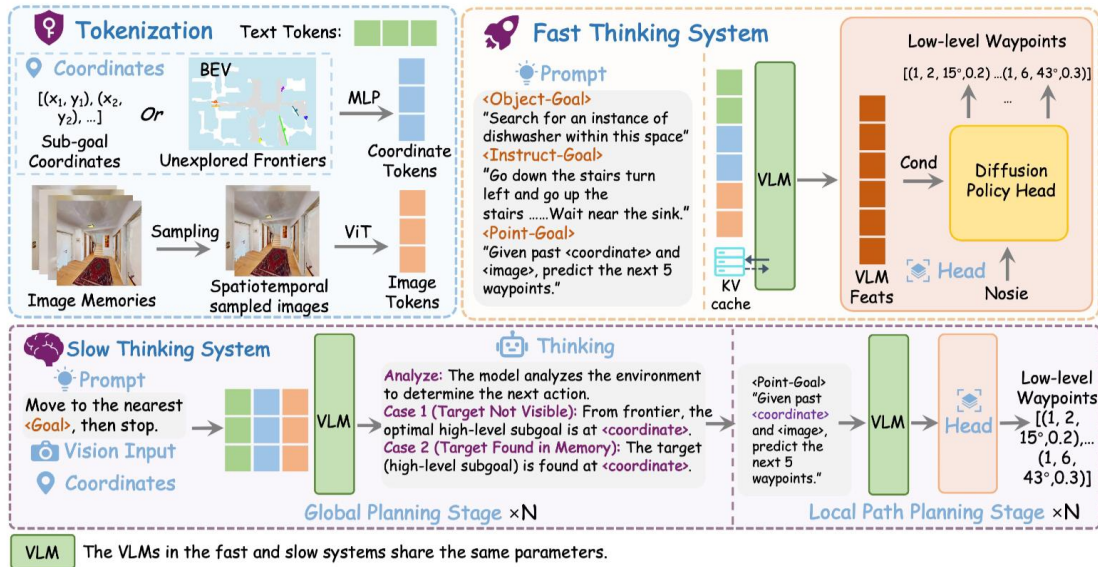
VLA in environments that

☆ 저장 50 인용 174회 인용



Visual-Language Navigation

- VLN: VLA for Navigation



Visual-Language Navigation

for Human-Robot Interactive Navigation



Instruction:

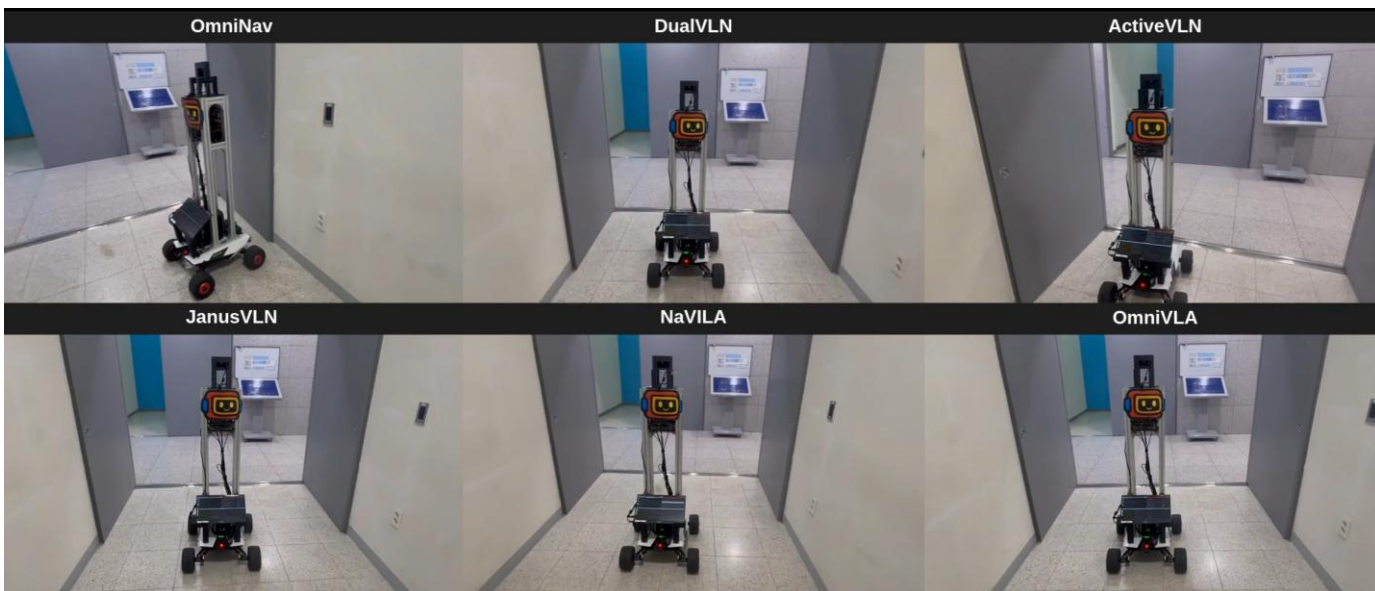
Walk forward along the white wall. Once you are close to the passage on your right, make a right turn and enter the passage. Continue walking straight, following the black striped tiles on the floor. After passing the fire hydrant, turn slightly to the right and move forward until you see a black sofa. Walk towards the black sofa and stop in front of it to finalize the trajectory.



Instruction: Walk forward along the white wall. Once you are close to the passage on your right, make a right turn and enter the passage. Continue walking straight, following the black striped tiles on the floor. After passing the fire hydrant, turn slightly to the right and move forward until you see a black sofa. Walk towards the black sofa and stop in front of it to finalize the trajectory.



Are We Ready for VLN in the Real World?



Instruction: Go straight ahead and turn right. Then keep going straight, and when you turn right at the corner, you'll see a blue door labeled 105 on your left. Stop in front of it.

Section 4

Multimodal Interaction (what is the human-like way?)



Visual-Language Nav meets HRI

- In the real world people don't describe every (low-level) detail.
 - They point.
- Point-to-Language Grounded Navigation via Diverse Map Interfaces



Scout Mini를 원하는 곳으로 불러주세요



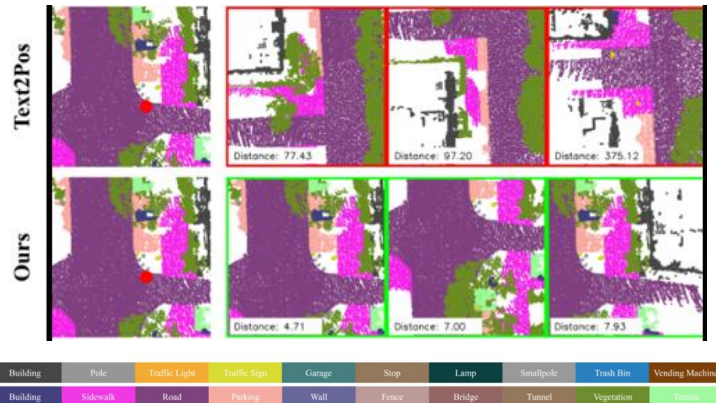


Visual-Language Nav meets HRI

- Language meets robot sensing modality
- Humans think in language; robots think in sensor data.
 - Language (Hint, **Human-side**) to Pointcloud (LiDAR map, **Robot-side**)

Query

- h_1 : The pose is on-top of a gray road.
- h_2 : The pose is on-top of a bright-gray smallpole.
- h_3 : The pose is south of a bright-gray smallpole.
- h_4 : The pose is west of a beige sidewalk.
- h_5 : The pose is south of a beige smallpole.
- h_6 : The pose is south of a gray-green pole.



Section 5

Reasoning or Geometry?

Visual-Language Nav, but why SLAM?

- Is Perception Foundation Model Still Necessary?

] 28 May 2026

Why Far Looks Up: Probing Spatial Representation in Vision-Language Models

Cheolhong Min¹, Jaeyun Jung¹, Daeun Lee¹, Hyeonseong Jeon¹,
Yu Su², Jonathan Tremblay³, Chan Hee Song^{3,†,‡}, and Jaesik Park^{1,†}



Luke Song • 2.

Research @NVIDIA Metropolis | Spatial Intelligence in VLMs

2 Tag(e) • Bearbeitet •



¹ Seoul National University

² The Ohio State University

³ NVIDIA



Do Vision-Language Models actually understand 3D space?

n, {cheolhong.min, jaesik.park}@snu.ac.kr

Do Vision-Language Models Understand 3D Scenes or Just Catalogue Objects?

[cs.CV] 19 May 2026

Visual-Language Nav, but why SLAM?

- Is Perception Foundation Model Still Necessary?

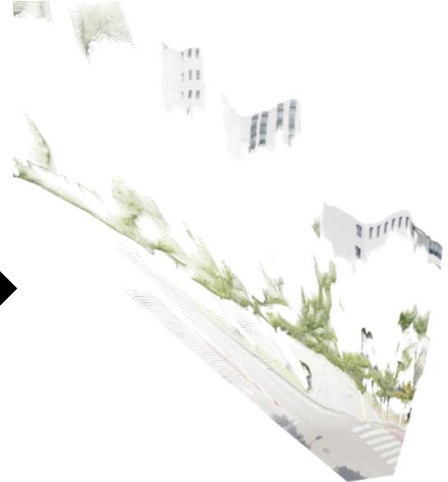
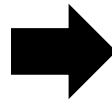
GPT 5.5 thinking



전방에 보이는 건물의 개수는?

몇 초 동안 생각함 >

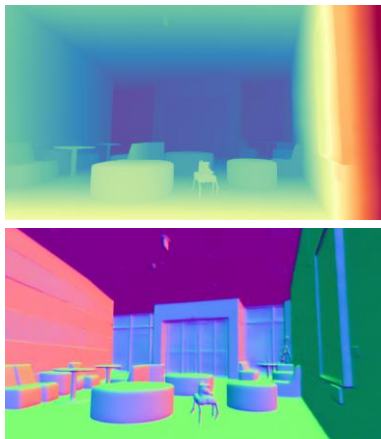
전방에 보이는 건물은 1개로 보여.



Wang, Ruicheng, et al. "Moge-2: Accurate monocular geometry with metric scale and sharp details." *Advances in Neural Information Processing Systems* 38 (2026): 35928–35959.

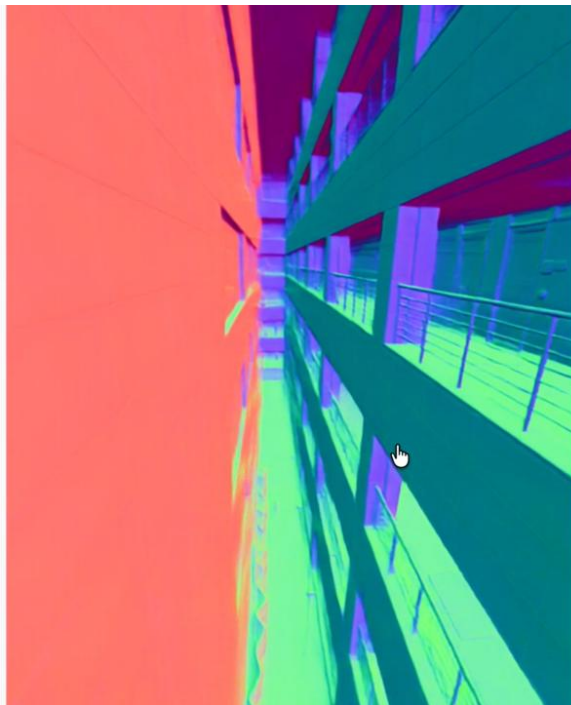
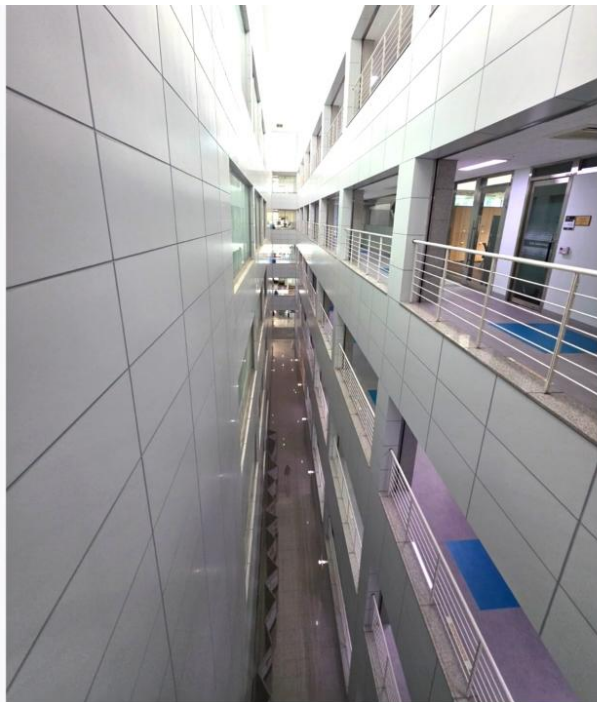
Visual-Language Nav, but why SLAM?

- Is Perception Foundation Model Still Necessary?



Visual-Language Nav, but why SLAM?

- Is Perception Foundation Model Still Necessary?



Visual-Language Nav, but why SLAM?

- Is Perception Foundation Model Still Necessary?
 - Multiple view consistency



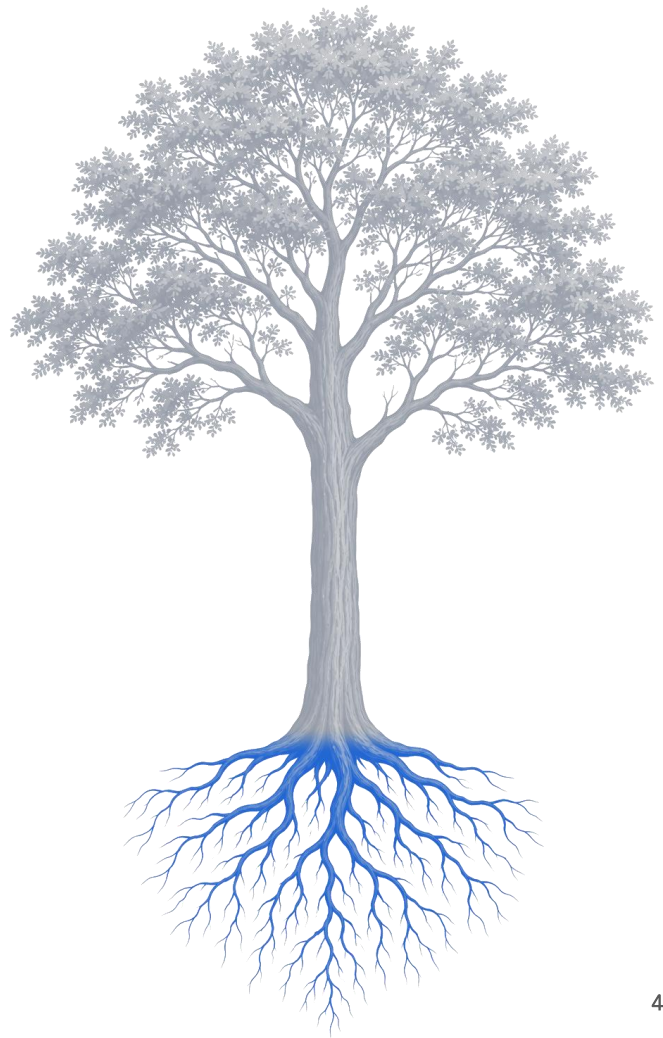
Visual-Language Nav, but why SLAM?

- Is Perception Foundation Model Still Necessary?
 - Multiple view consistency



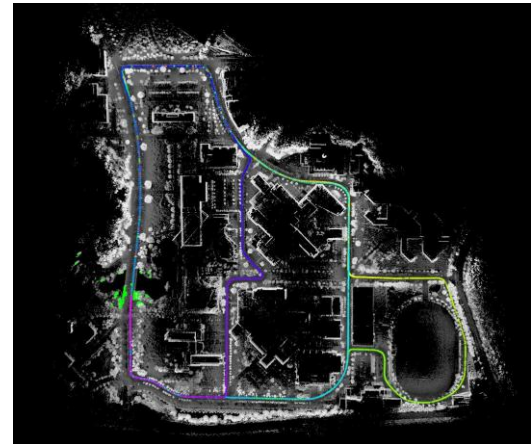
Zhang, Junyi, et al. "Logger: Long-context geometric reconstruction with hybrid memory." arXiv preprint arXiv:2603.03269 (2026).

Chapter 2: Past



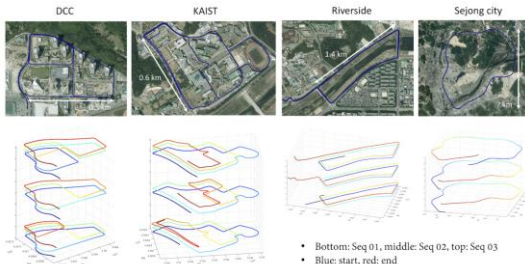
Scan Context (IROS18, TRO22)

- What? – Robust LiDAR Place Recognition
- Why? – 누구나 LiDAR Map을 손쉽게 만들 수 있게 하자.



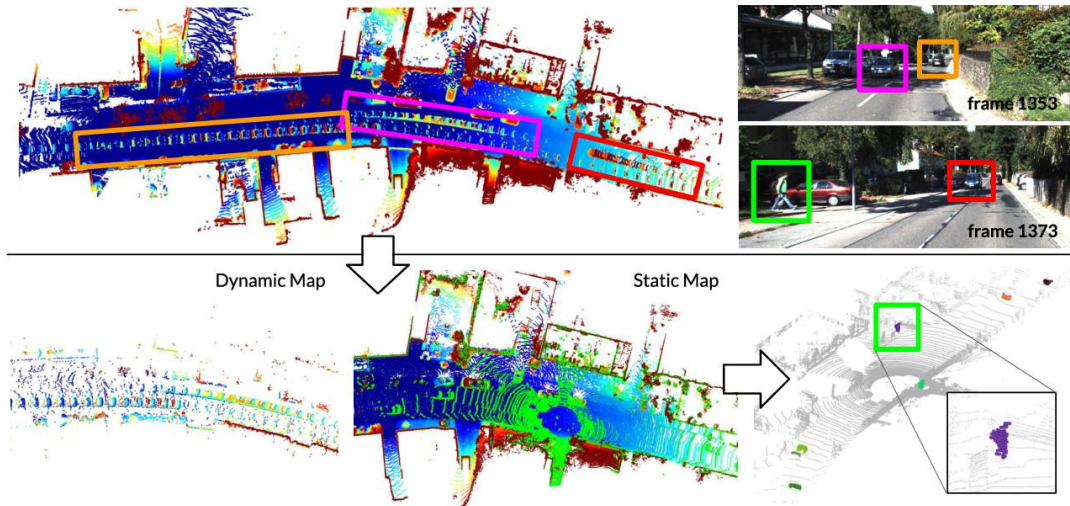
MuRan (ICRA20)

- What? – Beyond KITTI Dataset
- Why? – 누구나 LiDAR Map을 손쉽게 만들 수 있게 하려면 학계 전체가 진짜 문제를 풀게 해야 한다.



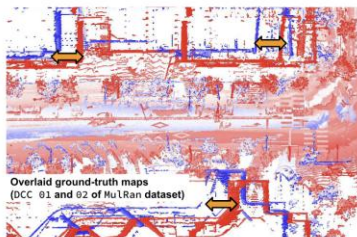
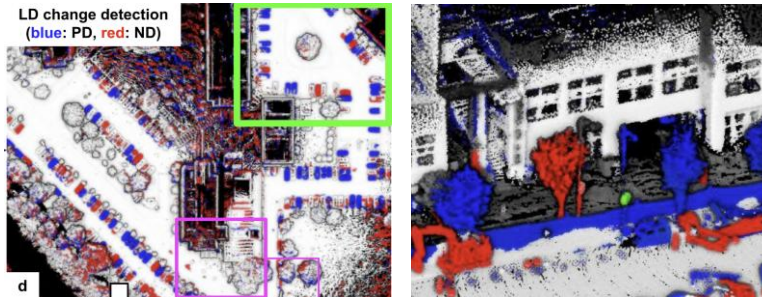
Removert (IROS20)

- What? – Make Clean Map
- Why? – 누구나 하위 어플리케이션에서 실제로 쓰고 싶을만한 LiDAR Map을 손쉽게 만들 수 있게 하자.

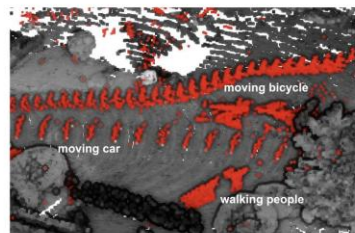


LT-mapper (ICRA22)

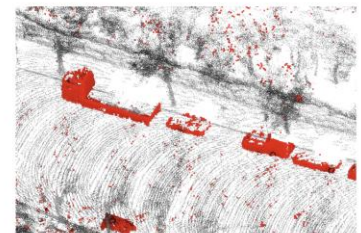
- What? – Long-term Map Management
- Why? – 세상은 변하고... (RTK GPS-INS 는 비싸다).



(a) Inconsistent ground-truths over sessions



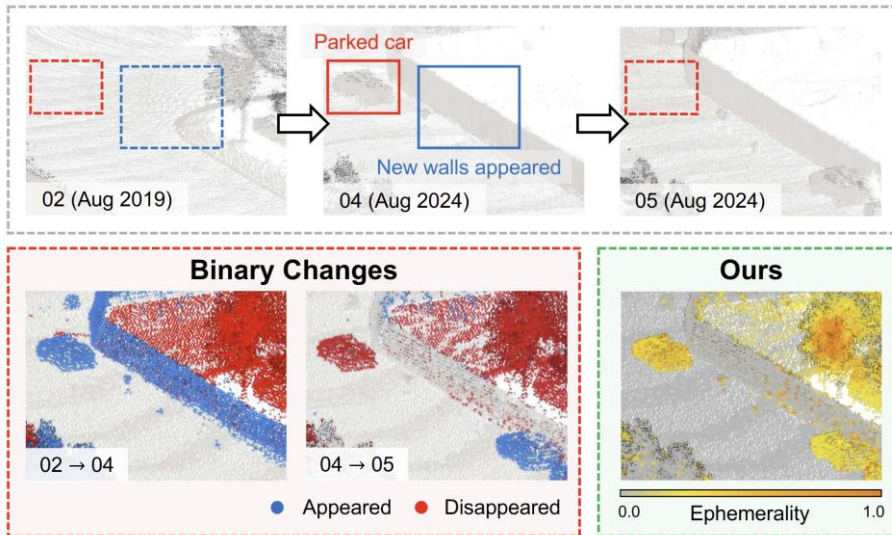
(b) Noisy points from moving objects



(c) Disappearing objects

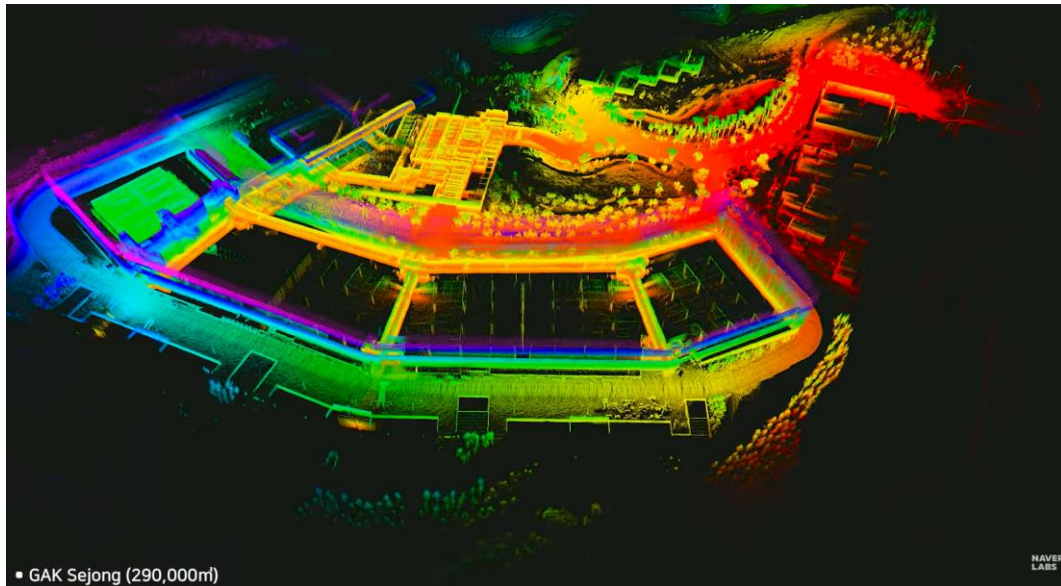
ELite (ICRA25)

- What? – Long-term Map Management (cont'd)
- Why? – 동적 변화에도 정도가 있다 (Ephemerality)



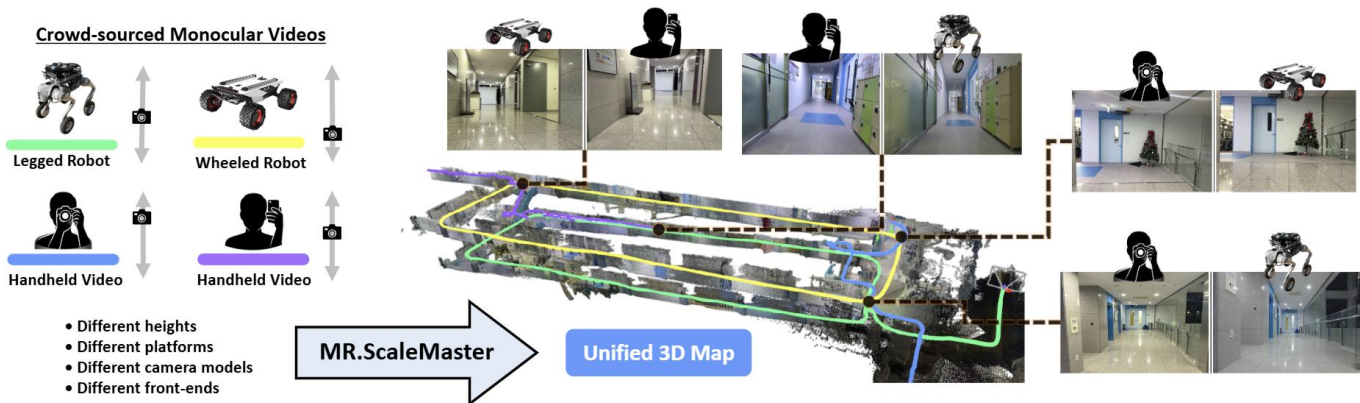
NAVER LABS (2022–2024)

- What? – 대규모 Mapping
- Why? – The Bitter Lessons



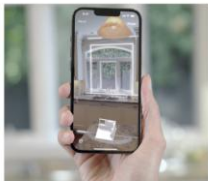
MR.ScaleMaster (IROS26-rejected)

- What? – Collaborative Mapping from Crowd-Sourced Videos
- Why? – 누구나 Map을 손쉽게 만들 수 있게 하자.



G. Kim Research Statement (2024)

- What? – Robot Web 시대의 정의와 연구주제
- Why? – Robot Intelligence 의 Flywheel Moment



Apple iPhone LiDAR RoomPlan



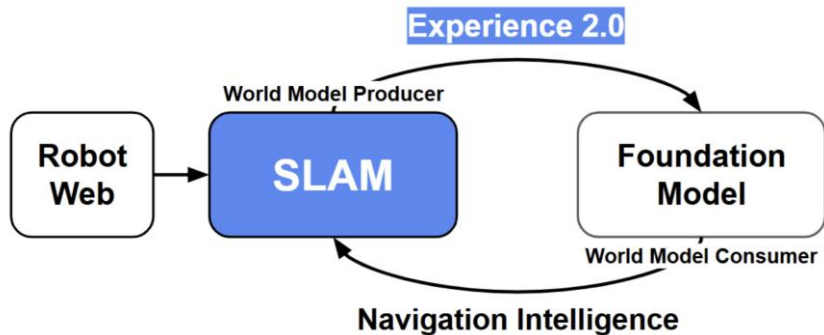
Project Aria from Meta



Unitree Go1 Legged Robot



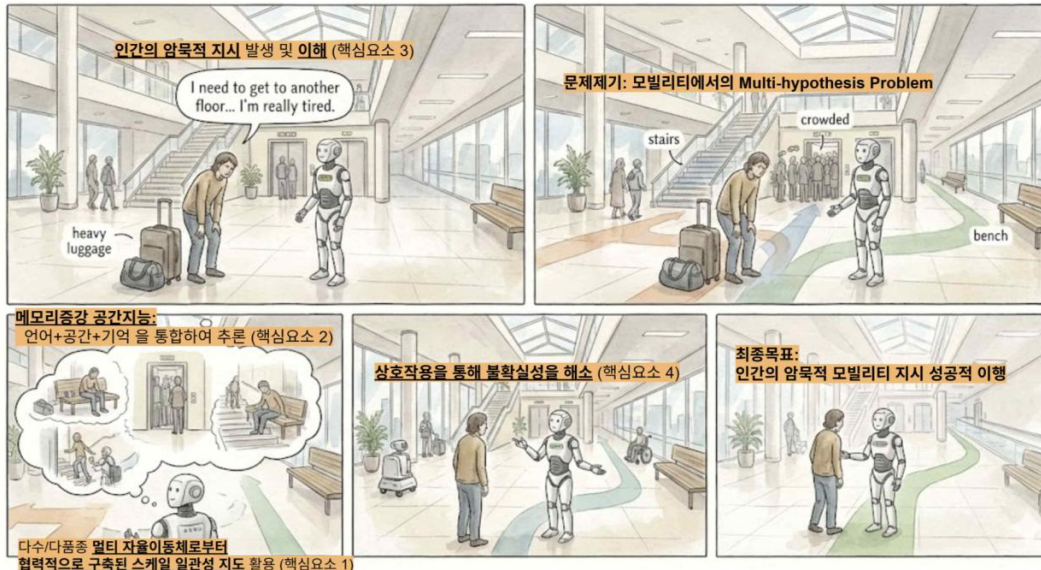
NAVER LABS Unmanned Shuttle (ALT-B)



LT-Mem (IROS26)

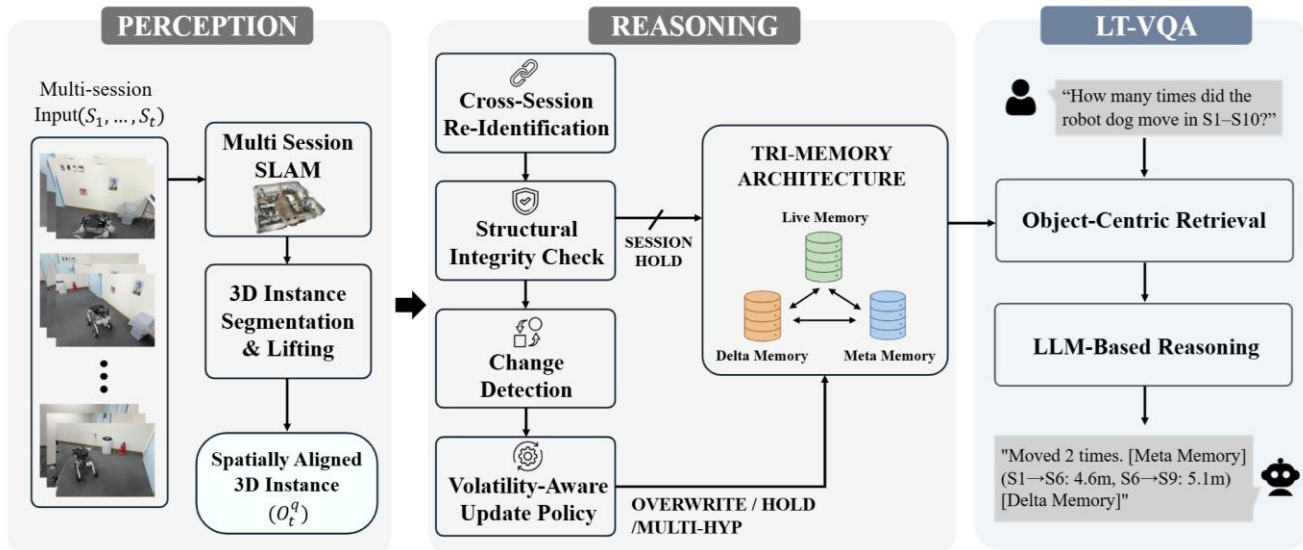
- What? – Long-term *Map* Memory Management
- Why? – 뭣이 중헌디

Memory-Augmented Spatial AI for Autonomous Mobility



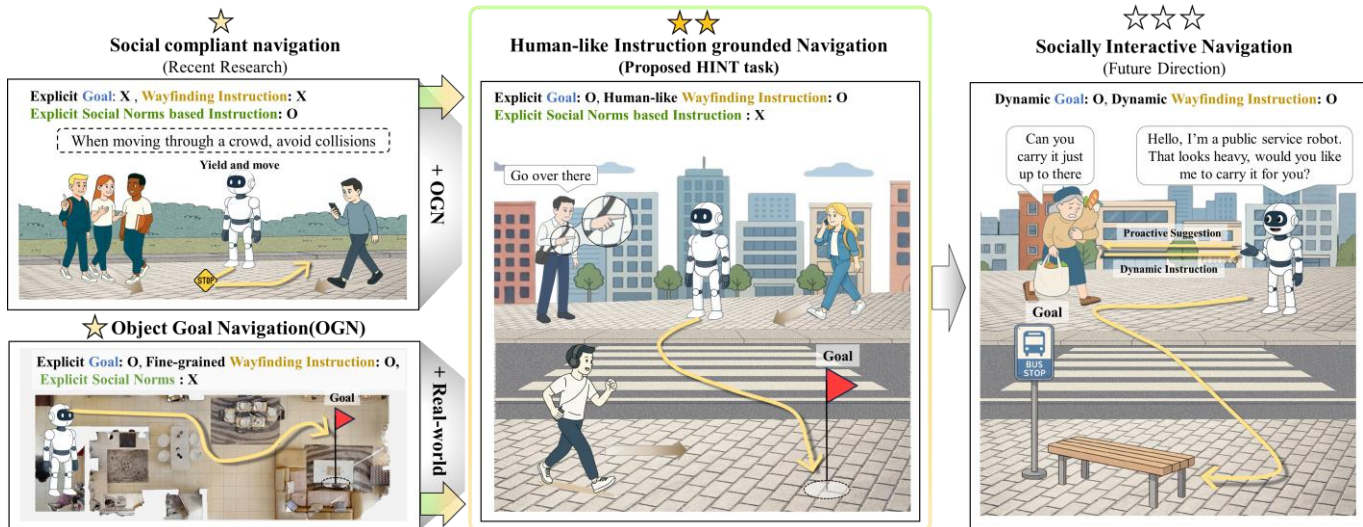
LT-Mem (IROS26)

- What? – Bridging the Perception–Reasoning Gap
- Why? – SLAM 과 VLM 이 만나는 그 어딘가. Map to Memory Moment.

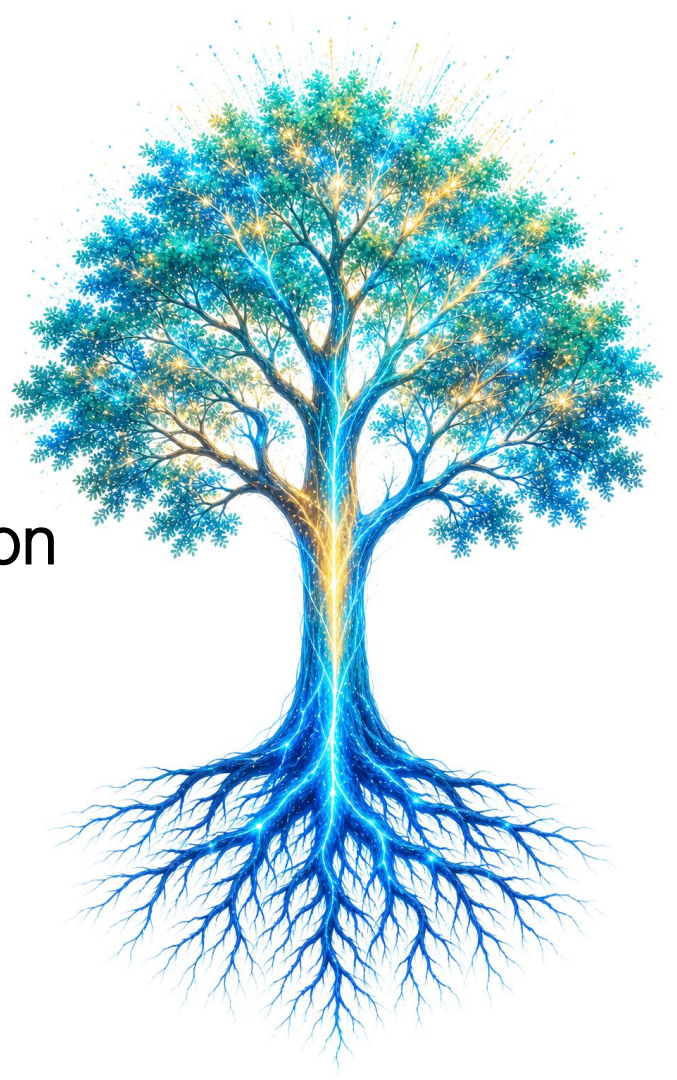


Human-robot Interactive Navigation

- 암묵적 인간 지시를 이해하고 상호작용하는
다중 자율 모빌리티를 위한 메모리 증강 공간지능 개발 (2026-2029)



Chapter 3: Conclusion



Summary

Future Prediction

- Number of robots ↑
- Number of human–robot interactions ↑ ↑ ↑

Problem

- Diversity of robots, sensors, and configurations ↑
- Complexity, errors, and mission failures ↑ ↑ ↑

Solution

- Bridging Geometry and Reasoning gap
- **Visual Language Multi-robot SLAM (APRL's Method)**

Summary

- Help robots understand human language better.
- **Human–robot Interactive Navigation (APRL's Task)**

APRL Vision

- ‘로봇 웹’ 시대는 온다.
- *‘어딜가나 로봇이’* 시대를 상상하고 함께하자.
- Human-robot Interactive Navigation

Closing: Language for Interaction

- How can this guy avoid getting fired? (이 친구가 팔리지 않으려면?)

